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## PROFESSIONAL SUMMARY:

B.E. Computer Science and Engineering student with hands-on experience building Machine Learning, NLP, and AI applications. Developed projects including a RAG-based AI portfolio assistant, earthquake risk prediction system, and handwritten character recognition model. Experienced in Python, Scikit-learn, NLP, embeddings, data preprocessing, model development, APIs, and deployment.

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## Technical Skills:

**Programming:** Python, SQL, C, Java

**Machine Learning:** Supervised Learning, Unsupervised Learning, Classification, Regression, Feature Engineering, Model Evaluation

**Deep Learning:** Neural Networks, CNNs, TensorFlow, Keras, PyTorch

**NLP & Generative AI:** Natural Language Processing, RAG (Retrieval-Augmented Generation), LLM Integration, Text Embeddings, Semantic Search, Prompt Engineering

**Computer Vision:** Image Processing, OpenCV, Handwritten Character Recognition, MNIST/EMNIST

**Data & Tools:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook, Git, MySQL

**Deployment & APIs:** Flask, REST APIs, API Integration

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## PROJECTS:

- **RAG-Based AI Portfolio Assistant** - Python, LLM, Vector Embeddings  
Built a retrieval-augmented AI assistant capable of answering questions using information from a personal knowledge base.
  - **Earthquake Risk Prediction for Nepal** - Python, Scikit-learn  
Developed a Random Forest model to classify regional earthquake risk using historical earthquake data and engineered geographic features.
  - **Handwritten Character Recognition** - Python, CNN, OpenCV  
Developed an image classification system for recognizing handwritten characters using image preprocessing and deep learning.
  - **Stock Market Prediction (NEPSE)** - Python, LSTM, Scikit-learn (**Currently Working**)  
Built a time-series prediction system using historical market data and technical indicators to predict stock returns and direction.
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## EDUCATION:

- Bachelor of Engineering (B.E.) – Computer Science and Engineering, Excel Engineering College, Anna University, Namakkal, Tamil Nadu, Year: **2024 – 2028**
  - Kanchan Secondary English School (KSES), Lalitpur, Nepal | **85%**
  - Shree Bajrabarahi Secondary School, Lalitpur, Nepal | **81.75%**
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## CERTIFICATIONS:

- Coursera: IBM Data Science Professional Certificate
- NPTEL: Programming in Java
- Great Learning: Graphic Design with Photoshop
- Great Learning: Introduction to R